

YUGESH BHOGE

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PUBLICATIONS

- 1. Constraining Equation of State of neutron star using neutron star-black hole mergers** Sept 1, 2026
Vasudev Dubey, Rahul Kashyap, Yugesh Bhoge
◦ [arXiv:2608.30995](#)
- 2. Distinguishing Kilonovae from Binary Neutron Star and Neutron Star-Black Hole Mergers** March 13, 2026
Ish Gupta, Yugesh Bhoge, Rahul Kashyap, Mukul Bhattacharya
◦ [arXiv:2603.13484](#) (In review at A&A)

RESEARCH PROJECTS

- 1. Bayesian formalism for distinguishing kilonova from NSBH and BNS mergers** Jul 2024 - March 2026
Guided by Prof Rahul Kashyap IIT Bombay
◦ Parameter estimation of intrinsic CBC parameters from lightcurves; ejecta mass & velocity → component masses & tidal deformability conversion; **EM-only CBC** parameter analysis through **next-generation detectors**; lifting merger-type degeneracy
- 2. Rayleigh-Taylor Instability in Stellar Interiors**  Jan 2025 - Apr 2025
Guided by Prof Rahul Kashyap IIT Bombay
◦ **General MHD dispersion relation** for RTI in stellar interiors; stability map plots; critical/fastest-growing wavelength prediction; validated via **2D finite volume simulations**; **magnetic-field stabilisation** confirmed (**shorter wavelengths**); RTI suppression conditions identified
- 3. Design and Fabrication of Suspension System for All-Terrain Vehicle** Mar 2019 - Apr 2020
Guided by Prof Sunil Damhare Pune University
◦ Design, analysis, fabrication & testing of off-road ATV suspension system; camber gain minimised to **0.0098°/mm** wheel travel; ride frequency **1 - 1.8 Hz (front), 1.1 - 2.0 Hz (rear)**; experimentally verified within **1.6%**
- 4. Study on Aerospike Engine and its Performance Comparison with Conventional Bell-Cone type Nozzle in Single Stage To Orbit Flight**  Mar 2019 - Nov 2019
Guided by Prof Sunil Damhare Pune University
◦ Aerospike vs. bell-cone nozzle comparison (CAD, CFD modelling); **PMF reduced 7.00%**; **thrust coefficient increased 26.66%**; enables SSTO feasibility, reduced fuel, increased payload capacity.

OTHER PROJECTS

- 1. Blindspots in omnidirectional Interferometric Gravitational Waves detectors** Oct 2024 - Dec 2024
Tools: PyCBC 
◦ Directional sensitivity/blind-spot analysis for **LIGO, Virgo, KAGRA** using **PyCBC**; antenna pattern function computation; developed and tested multi-detector network mitigation strategies
- 2. Demonstration of chaos by period doubling route using Malkus waterwheel** Aug 2024 - Oct 2024
Tools: Python, ImageJ 
◦ Malkus waterwheel construction + mathematical model (**Lorenz system equation, limit cycle, pitchfork, subcritical Hopf bifurcation**); chaotic behaviour under steady flow; **self-written Python code**; **ImageJ** image analysis & tracing
- 3. Transforming Alphabets into Touch: A Braille Language Project** Sep 2023 - Dec 2023
Tools: Arduino UNO, Arduino IDE, C++ 
◦ **26-letter Braille** → **6-segment display** mapping; null-character time-delay pause for word/sentence distinction; 6-servo haptic feedback display; **Arduino UNO** control
- 4. The Eco Coach** May 2018 - Dec 2019
Design Project
◦ Indian Railways design project – plastic litter mitigation; integrated shredder + storage assembly for in-coach plastic waste

EDUCATION

- Indian Institute of Technology, Bombay** Jul 2023 - May 2025
MSc Physics Mumbai, India
◦ CPI: 7.54/10.00
- Savitribai Phule Pune University** Aug 2016 - Apr 2020
BTech Mechanical Engineering Pune, India
◦ CGPA: 8.85/10.00

PROFESSIONAL TEACHING AND RESEARCH EXPERIENCE

- **Project Associate (Remote), Numerical Relativity Group** May 2025 – Present
Numerical simulation & statistical inference IIT Bombay
 - Kilonova population analysis; paper submitted to A&A (arXiv:2603.13484, Mar 2026, under review); NSBH EoS constraints pipeline (arXiv:2608.30995); presented YAM 2026; active LIGO Scientific Collaboration member.
- **Gen AI expert** Jan 2026–Present
 - Training AI models for experimental outcome prediction of physical experimentations Handshake.AI, Outlier.AI
- **Freelance Educator & Independent Professional** Dec 2020 – Jul 2023
 - Corporate role (Dec 2020 – Apr 2021); freelance private tutoring – physics & mathematics (high school, undergraduate); JAM 2023 preparation, qualified **99.45th percentile**

TALKS AND WORKSHOPS

- **The Young Astronomers Meeting 2026** Mar 2026
Hybrid CAMK, Poland
 - 3-day international meeting – young & veteran collaborators worldwide; presented recently published work
- **GW - Detector Characterization Workshop** Dec 2025
LISC IUCAA, Pune, India
 - Hands-on training – noise subtraction, glitch characterisation, spectral analysis, ML applications in LIGO data analysis

HONORS AND AWARDS

- **Joint Admission for Masters** May 2023
 - **99.45th percentile** – nationwide masters entrance examination
- **INSPIRE Scholarship** May 2016
Department of Science & Technology, Government of India
 - **Top 1%** nationally – Class 12 academic excellence

PROFESSIONAL MEMBERSHIPS

- **LIGO, LSC and LISC**, Membership ID: yugeshjivaram.bhoge@ligo.org Feb 2025 - Present

SKILLS

- **Computational Astrophysics:** NumPy, SciPy, Astropy, Bilby, LALSuite, PyCBC, MOSFiT
- **Simulation & Engineering:** FVM, CFD (Ansys), CAD (SolidWorks), Arduino IDE, LTSpice, ImageJ
- **Mathematical & Statistical Tools:** MATHEMATICA, L^AT_EX, OriginLabs, MATLAB, Scilab
- **Programming & HPC:** Python, C++, Bash, HTML, HTCondor, Cuda, Cray

LEADERSHIP EXPERIENCE

- **Internship Coordinator** Aug 2023 - May 2024
Training and Placement Cell, IIT Bombay 
 - Streamlined internship processes for 1500+ students; collaborated with 300+ firms globally, designed targeted plans for diverse student interests in core engineering, finance, and research
- **Team Captain** Jan 2019 - Apr 2020
Genesis 16 Motorsports, DYPIEMR, SPPU 
 - Managed a team of **25-30** people directly, and worked on building an **off-road racing vehicle**, signed multiple **sponsorship deals worth Rs.50,000** with different industries for the project, convinced the institute for the budget approval of **Rs 2,50,000**

VOLUNTEER AND OUTREACH EXPERIENCE

- **PE ROTA volunteer** Aug 2025
LIGO Scientific Collaboration
 - 1 manual PE run launched; vanilla BBH event, PE ROTA shift
- **Scientific summary translator** Jul 2025
LIGO Scientific Collaboration
 - **GW231123** science summary translation – Marathi
- **LIGO India Outreach Volunteer** Dec 2024 & Dec 2025
LIGO INDIA 
 - TechConnect (Techfest) outreach, 8-peer team; GW awareness & misconception dispelling; activities: “Black Hole Ping Pong” game, “Stretch and Squash” selfie tool

CERTIFICATIONS

- **International Centre for Theoretical Science, Bangalore:** Fascinating worlds of flows *Jul 2023*
- **Dassault's Systemes:** Certified SolidWorks Professional *Dec 2020*
- **Coursera & MathWorks:** Exploratory Data Analysis with MATLAB *Jul 2020*

REFEREE DETAILS

1. Prof Dr Rahul Kashyap

Designation: Assistant Professor

Affiliation: Indian Institute of Technology, Bombay

Relation: Thesis advisor and mentor

Email ID: rahulkashyap@iitb.ac.in

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2. Dr Ish Mohan Gupta

Designation: Postdoctoral Fellow

Affiliation: University of California, Berkeley and Northwestern University

Relation: Mentor and project collaborator

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